Course Code	**FLUID MECHANICS**	L	T	P	C
23A01506	**HYDRAULIC MACHINES LAB**	0	0	0	1.5

Course Objectives:

The objectives of this course are to make the student to:

1. **Understand** the principles of fluid mechanics and validate fundamental concepts through experiments.
2. **Determine** discharge coefficients for various flow measurement devices and analyze flow behavior.
3. **Evaluate** energy lossesin pipes, open channels, and hydraulic jumps to improve flow efficiency.
4. **Analyze** the impact of jet forces on vanes and their applicationsin hydraulic machinery.
5. **Assess** the performance characteristics of hydraulic turbines and pumps under different operating conditions.

Course Outcomes:

Upon successful completion of this course, students will be able to:

1. **Verify** Bernoulli's equation and apply it to real-life fluid flow problems.
2. **Determine** the coefficient of discharge for orifices, notches, and flow meters.
3. **Evaluate** head losses due to friction and minor lossesin pipe flow systems.
4. **Analyze** the impact of jets on vanes and its significancein hydraulic machinery.
5. **Assess** the performance of turbines and pumps under different conditions and recommend optimal operating parameters.

CO – PO Articulation Matrix:-

Course Outcomes	**PO 1**	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**	**PO 12**	**PSO1**	**PSO2**
CO -1	3	2	2	2	-	-	-	-	-	2	-	2	3	2
CO -2	3	3	3	2	2	-	-	-	-	-	-	1	3	2
CO -3	3	3	2	2	2	-	-	-	-	-	-	1	3	3
CO -4	3	3	3	2	2	-	-	-	-	-	-	1	3	3
CO -5	2	2	2	1	3	-	-	-	-	2	1	2	2	2

List of Experiments

1. Verification of Bernoulli's Equation
2. Determination of Coefficient of Discharge for A Small Orifice By A Constant Head Method
3. Calibration of Venturimeter/ Orifice Meter
4. Calibration of Triangular / Rectangular/Trapezoidal Notch
5. Determination of Minor Losses in Pipe Flow
6. Determination of Friction Factor of A Pipeline
7. Determination of Energy Loss in Hydraulic Jump
8. Determination of Manning's andChezy's Constants for Open Channel Flow.
9. Impact of Jet On Vanes
10. Performance Characteristics ofPelton Wheel Turbine
11. Performance Characteristics of Francis Turbine
12. Performance Characteristics of Kaplan Turbine

13. Performance Characteristics of A Single Stage / Multistage Centrifugal Pump

Note: Minimum 10 out of the above are to be conducted.

TEXT BOOKS:

1. Desmukh T. S., A lab manual on Fluid Mechanics and Hydraulic Machines, Laxmi Publications
2. Dr. S.K. Panigrahi, Ms. L. Mohanty, Fluid Mechanics and Hydraulic Machines Laboratory Manual ,S.K.KATARIA&SONS ,Educational Publisher.

REFRENCE BOOKS:

1. Dr. N. Kumara Swamy, Fluid Mechanics and Machinery Laboratory Manual, Chartor Publications
2. D. Sathish, Fluid Mechanics and Machinery Lab Manual, BP International Publications

Online Learning Resources:

https://archive.nptel.ac.in/courses/112/106/112106311/